(ii) From the graph, the required number of days to complete the work when the company decides to work with 120 workers is 50 days.

Also, from $xy = 6000$ if $x = 120$, then $y = \frac{6000}{120} = 50$

(iii) From the graph, if the work has to be completed by 200 days, the number of workers required is 30.

Also, from $xy = 6000$ if $y = 200$, then $x = \frac{6000}{200} = 30$

Example 3.50

Nishanth is the winner in a Marathon race of 12 km distance. He ran at the uniform speed of 12 km/hr and reached the destination in 1 hour. He was followed by Aradhana, Jeyanth, Sathya and Swetha with their respective speed of 6 km/hr, 4 km/hr, 3 km/hr and 2 km/hr. And, they covered the distance in 2 hrs, 3 hrs, 4 hrs and 6 hours respectively.

Draw the speed-time graph and use it to find the time taken to Kaushik with his speed of 2.4 km/hr.

Solution: Let us form the table with the given details.

Speed x (km/hr)	12	6	4	3	2
Time y (hours)	1	2	3	4	6

From the table, we observe that as x decreases, y increases. Hence, the type is **inverse variation.**

Let $y = \frac{k}{x}$

$\Rightarrow xy = k$, $k > 0$ is called the constant of variation.

From the table $k = 12 \times 1 = 6 \times 2 = \ldots$

$= 2 \times 6 = 12$

Therefore, $xy = 12$.

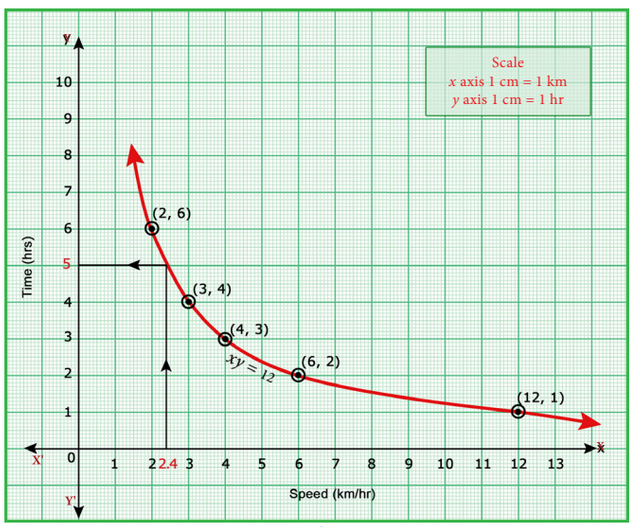


Fig. 3.14

Plot the points (12,1), (6,2), (4,3), (3,4), (2,6) and join these points by a smooth curve (Rectangular Hyperbola).

From the graph, we observe that Kaushik takes 5 hrs with a speed of 2.4 km/hr.

Note

Already we learned that, the linear equation of straight line is $y = mx+c$, where m is the slope of the straight line and c is the y – intercept. Also, the equation reduces to $y = mx$ when the straight line passes through origin. As the graph of direct variation refer to straight line and its general form is $y = kx$, we can conclude that '**constant of proportionality**' is nothing but '**slope**' of its straight line.

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